

Bachelor in Computer Applications

Course Title: **Operating System**

Code No: Semester: **3**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

1. Suppose that a disk has 100 cylinders, numbered 0 to 99. The drive is currently serving a request at cylinder 45. The queue of pending request, in FIFO order is: 88, 72, 13, 74, 48, 9, 22, 50, 35 and 50. What is the total distance (in cylinders) that the disk arm moves to satisfy all pending request for each of the following disk scheduling algorithms? a) FCFS b) SCAN c) SSTF
2. Why virtual memory technique is used in the computer system? What is logical address? Explain the process of conversion of logical address to physical address in single level paging scheme.
3. What is critical section? What are the operations that can be performed on the semaphore? Explain the solution to producer consumer problem using semaphore.
4. Draw and describe the 5-state process model.
5. What is kernel? Differentiate between monolithic and microkernel structure.
6. What is DMA? Explain how it works in brief with suitable diagram.
7. What are different types of file? Describe the sequential and random access of files.
8. What is deadlock? What are necessary conditions for deadlock? Explain.
9. What is thread? Explain user level thread and kernel level thread.
10. A 2 GB hard disk has 2 KB block size, calculate the size of bitmap for bitmap based free space management.
11. What is an operating system? Differentiate between time sharing and real time operating system.
12. Consider the following set of processes, with the length of CPU burst time given in milli seconds.

Process	CPU Burst (ms)	Priority	Arrival Time (ms)
A	83	0	B91
B	91	(Lowest)	0C10
C	20	D64	(Highest)
D	64	0	